

Landing Profile and Powered Descent Strategy for Chandrayaan-3

**Aditya Rallapalli¹, Suraj Kumar¹, GVP Bharat Kumar¹, Chiranjib Guha Majumder²,
Debajyoti Chakrabarti², MP Rijesh²**

¹Mission Simulation Group, U.R. Rao Satellite Centre, Bangalore, India

²Control Dynamics and Design Group, U.R. Rao Satellite Centre, Bangalore, India

Abstract

Chandrayaan-3 became the first lunar robotic exploration mission to accomplish soft touchdown near lunar south pole. This was achieved through years of design, development and rigorous testing at sub-system followed by system level integrated tests. Simulations were carried out to validate the design of the powered descent profile which evolved through extensive iterations to satisfy all the mission constraints. These constraints include sensors and actuator operating limits, optimizing between tolerable navigation error and guidance convergence limits, system ability to handle multiple failure scenarios and sufficient robustness in design. This paper presents the landing strategy and powered descent profile designed for Chandrayaan-3.

Section I Introduction

Chandrayaan-3 is configured as a two-module system comprising of a Propulsion Module (PM) and a Lander Module (LM). A six-wheel Rover is accommodated inside the Lander Module to perform lunar exploration post landing. On 14th July 2023, Chandrayaan-3 integrated module was launched in the Earth parking orbit (180 km x 36000 km) by GSLV MK-III. Propulsion module was used to raise the apogee of Earth parking orbit, perform trans lunar insertion (TLI) and lunar bound maneuver to place the composite module into 100 km lunar circular orbit. Lander module got separated from propulsion module in 100 km lunar orbit. Newly developed throttling engines (800 [N]) were calibrated while performing deorbit maneuver exercised to attain elliptical orbit around moon (30 km x 100 km) designed for powered descent. On 23rd August 2023, the lander initiated autonomous powered descent at an altitude of 30 km and approximately 750 km from preselected landing site using four throttleable liquid engines. Navigation, Guidance and Control system steered the lander through multiple phases to perform soft touchdown on lunar surface. Chandrayaan-3 became the first Indian lunar landing mission to soft and safe land near south pole. It landed to a final latitude and longitude of 69.37356° S and 32.31975° E which was approximately 300 m away from the initial landing point preselected based on landing site analysis.

Powered descent phase comprises of multiple sub-phases which ensures soft and safe landing. These sub-phases comprise of rough braking, attitude hold, fine braking and terminal descent phase. Terminal descent phase consists of Hovering at higher altitude, vertical descent, hazard avoidance followed by constant velocity descent. Entire powered descent profile for Chandrayaan-3 lasted for approximately 1140 s ensuring soft landing on moon.

This paper is organized as follows, section II introduces about Chandrayaan-3 mission objectives, configuration and powered descent phases, section III summarizes the constraints and studies done to design milestones of each phases. Section IV discuss about powered descent strategy multi-mission path and robustness introduced in Chandrayaan-3 powered descent design. Section V shows the flight performance of systems along with pre-flight predictions and bounds. Conclusions is given in section VI.

Section II Chandrayaan-3 Mission Objectives and Configuration

Primary mission objective of Chandrayaan-3 was to demonstrate, “Safe, soft and precise” landing near to lunar south pole. Deploy rover & Demonstrate its mobility on lunar surface for conducting in-situ scientific experiments. Lunar south pole region was chosen as pre-selected landing site as it has lots of scientific value. Landing co-ordinate was selected based multiple constraints such as terrain variation, illumination condition, communication to earth, sun elevation for power generation and local terrain features. Landing site identification was done using Chandrayaan-2 orbiter high resolution camera. Table 1 summarizes the two-landing co-ordinate which was selected based on constraints.

Table 1 Landing Co-ordinate selected for chandrayaan-3 mission

Landing Site	Landing Opportunity	[Latitude, Longitude] deg
Prime Landing Site (PLS)	23 rd August 2023	69.3676 ⁰ S 32.3481 ⁰ E
Alternate Landing Site (ALS)	26 th August 2023	69.4977 ⁰ S 17.3304 ⁰ W
Shiv Shakti Point (Ch-3 Touch down location)	23 rd August 2023	69.37356 ⁰ S 32.31975 ⁰ E [170 S, 300 E] m wrt to PLS

Configuration of Chandrayaan-3 was selected to ensure mission requirements are met in terms of landing site terrain variation, illumination condition, propellant margins, terminal conditions, redundancies etc. Table 2 provides details of sensor and actuator configuration selected for Chandrayaan-3. Figure 1 shows the 3D view and axis definition of Chandrayaan-3 in integrated configuration.

Table 2 Sensor and Actuator Configuration of Chandrayaan-3

Sensor	
Sensor	Functionality
Laser Altimeter (LASA)	Range along line of sight
Ka Band Radio Altimeter (KaRa)	Altitude from surface
Laser Doppler Velocimeter (LDV)	3-axis body velocity
Lander Horizontal Velocity Camera (LHVC)	Pixel Shift to compute velocity
Lander Hazard Detection and Collision Avoidance Camera (LHDAC)	Distance to safe landing site
Lander Position Detection Camera (LPDC)	Lander Latitude and Longitude
Touchdown sensor	Contact signal of lander and lunar surface
Actuator	
Actuator	Functionality
Throttleable Liquid Engines (800 [N] x 4)	Velocity reduction
Reaction Control System (58 [N] x 8)	Attitude Control

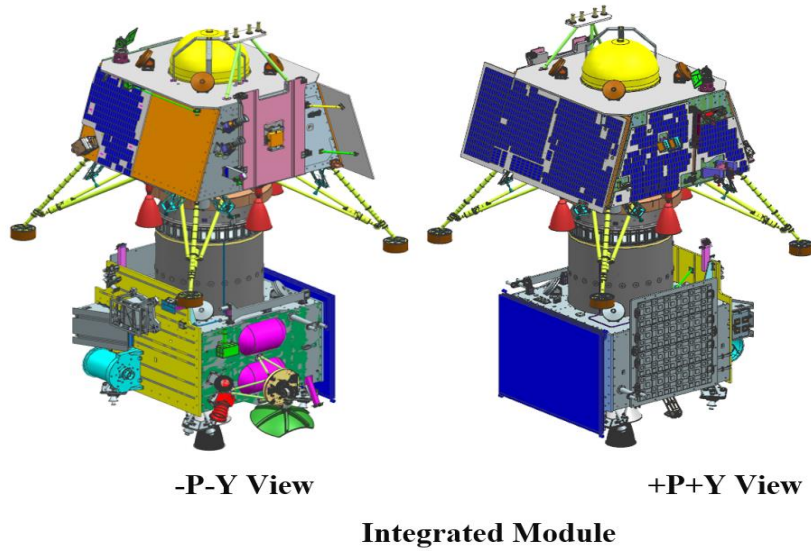


Figure 1 3D view of integrated Chandrayaan-3 configuration

Entire mission process from lift off till landing was divided into multiple phases which includes lift-off from earth, earth bound manoeuvres, trans lunar injection, lunar orbit insertion, lunar bound manoeuvres, deorbit and powered descent followed by surface exploration using rover. Figure 2 shows the entire mission profile.

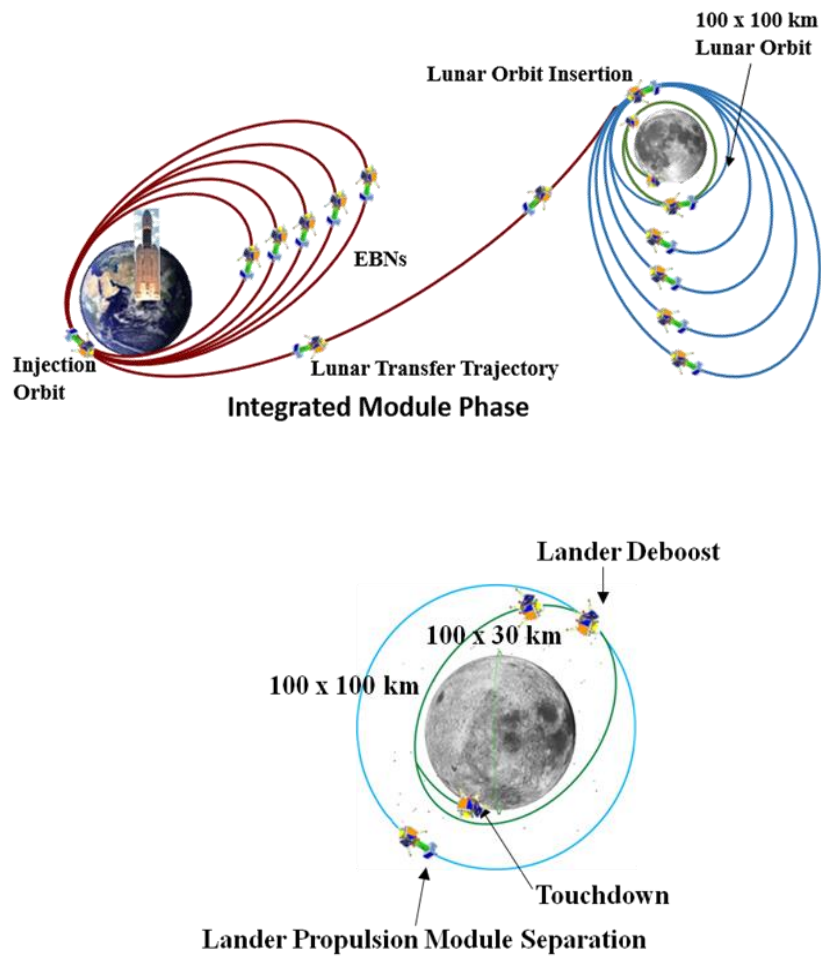


Figure 2 Mission profile of chandrayaan-3

Section III Chandrayaan-3 Powered Descent Strategy

Powered Descent in Chandrayaan-3 had multiple phases as shown in Fig.3. It starts with Rough Braking phase which extends from 30km to 7.4km. Most of the initial velocity ($\sim 1.68\text{km/s}$) is brought down in this phase by firing four 800N engines. Final conditions have to be achieved with thrust per engine as 740N and angle with respect to vertical as 50 degrees. This is followed by Attitude Hold phase of 10 seconds where there is no closed loop guidance but thrust and angle with respect to vertical are maintained. Attitude Hold ends with a height update from absolute sensor which is followed by a Fine Braking phase. The guidance algorithm here has to start from the last thrust and attitude command ensuring continuity such that the Lander reaches 800m height above the designated landing site with zero velocity. Upon reaching this height, at this juncture two engines are switched off and the Lander enters hovering for a duration of 12 seconds. Hovering guidance basically ensures that the velocities are near zero. At the end of hovering at 800m, height and velocities are updated by absolute sensors. This is followed by vertical descent to 150m with target velocities as zero. Second hovering starts at 150m for a maximum duration of 22 seconds wherein hazard camera is operated to give hazard free coordinate from current location and also with height and velocity updates from absolute sensors. This is followed by Re-Targeting to hazard free site but to a height of 60m with final velocity target as zero. After Re-Targeting, lander descends to 10m with target vertical velocity as -1m/s which is followed by constant velocity descent till touchdown.

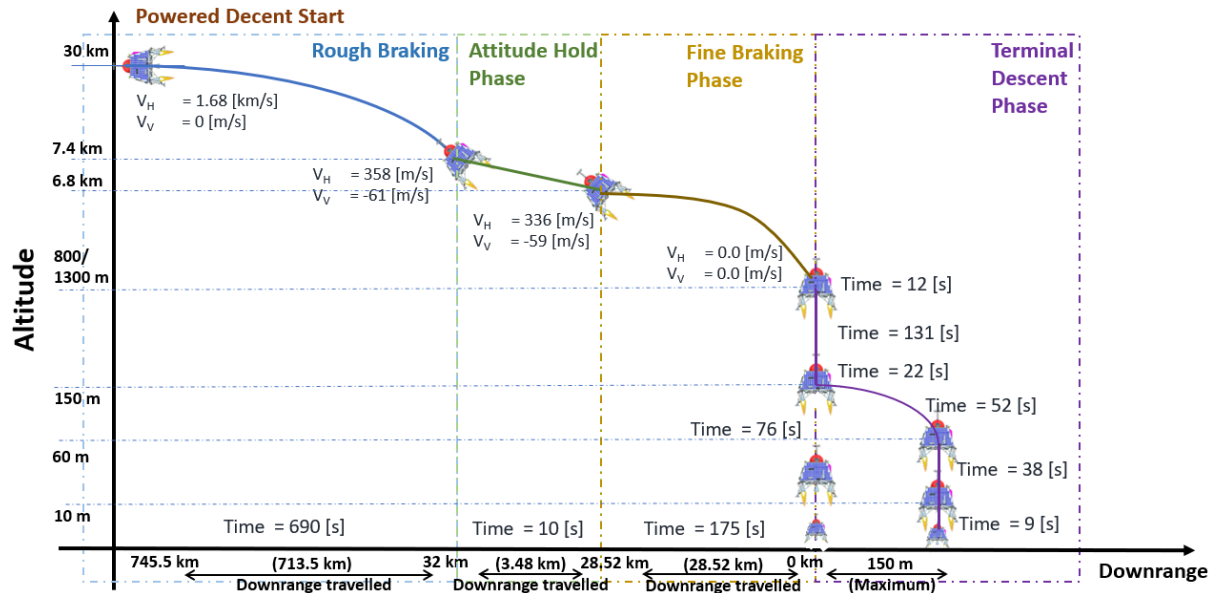


Figure 3 Powered Descent Profile for Chandrayaan-3 lander

Section IV Chandrayaan-3 Powered Descent Milestone Design

Rough Braking Phase

Objective of rough braking phase is to kill most of the initial velocity and travel maximum downrange towards landing site. Rough braking phase is initiated when lander is at perilune with an altitude of 30 km. Lander initial velocity is close to 1.688 km/s and is approximately 750 km from landing site. Table 3 shows the detailed rough braking phase parameters

Table 3 Rough Braking Phase Parameters

Phase Parameters	
Duration, [s]	690 s

Guidance	2 nd Order polynomial Close loop guidance, thrust regulation, Final Thrust constrained at 740N (Close to natural solution)			
Terminal constraint	Pitch Angle		50 ⁰	
	Velocity (Vh & Vv)		358 [m/s] & -61 [m/s]	
	Position (Dr & Alt.)		32 [km] & 7.481 [km]	
Initial Condition	Pitch Angle		Unconstrained	
	Velocity (Vh & Vv)		1680 [m/s] & 0 [m/s]	
	Position (Dr & Alt.)		745.5 [km] & 29.23 [km]	
Navigation	Pure Inertial Navigation (LIRAP Propagated)			
Initial OD Dispersion	Radial, [m]	22	Radial, [m/s]	0.375
	Along, [m]	312.2	Along, [m/s]	0.02
	Across, [m]	9.6	Across, [m/s]	0.006
RB End [#] Navigation Dispersion	Radial, [m]	830.365	Radial, [m/s]	1.678
	Along, [m]	576.074	Along, [m/s]	0.581
	Across, [m]	388.069	Across, [m/s]	1.126
Initial Guidance Execution Error	Radial, [m]	1000	Radial, [m/s]	2
	Along, [m]	0	Along, [m/s]	0
	Across, [m]	200	Across, [m/s]	0.2
Terminal Guidance Execution Error	Radial, [m]	27	Radial, [m/s]	0.2
	Along, [m]	43	Along, [m/s]	0.3
	Across, [m]	10	Across, [m/s]	0.15

Design of initial downrange to target and initial altitude is done based on ground trajectory optimization with a thrust constraint of 750 [N] x 4. Target milestone is selected considering following design considerations

- Downrange covered should be maximized (to ensure minimum DEM variation from landing site).
- Inertial navigation propagation error should be within correctable limits.
- At RB end, system parameter should be within sensor operational limits

Trajectory has a natural pitch angle of 50⁰ for the specified boundary condition, due to which sensors are mounted at an angle of 50⁰ with respect to -roll axis ensuring they point in nadir direction at the end of attitude hold phase.

Attitude Hold Phase

Main objective of attitude hold phase is aid the inertial navigation system using altimeter, thus correcting the navigation error. Altimeters used in this phase are LASA and KaRa which are mounted in such a way that, at the end of rough braking phase when lander pitch angle is 50⁰ both the altimeter look in nadir direction. This attitude needs to be maintained in order to get proper altimeter updates. To

ensure attitude is maintained, attitude control system is decoupled from guidance. Table 4 shows the detailed attitude hold phase parameters

Table 4 Attitude Hold Phase Parameters

Phase Parameters					
Duration, [s]	10 s				
Guidance	No Close Loop Guidance, Thrust regulation (4 x 740 [N])				
Terminal Condition	Pitch Angle		50 ⁰		
	Velocity (Vh & Vv)		336 [m/s] & -59 [m/s]		
	Position (Dr & Alt.)		28.52 [km] & 6.88 [km]		
Initial Condition	Pitch Angle		50 ⁰		
	Velocity (Vh & Vv)		358 [m/s] & -61 [m/s]		
	Position (Dr & Alt.)		32 [km] & 7.48 [km]		
Navigation	Pure Inertial Navigation (LIRAP Propagated) & Filter in background				
Initial AH# Dispersion	Radial, [m]	830.365		Radial, [m/s]	1.678
	Along, [m]	576.074		Along, [m/s]	0.581
	Across, [m]	388.069		Across, [m/s]	1.126
AH End# Dispersion	Radial, [m]	75*, 847** (-Ve direction)		Radial, [m/s]	1.711
	Along, [m]	590.927		Along, [m/s]	0.587
	Across, [m]	399.502		Across, [m/s]	1.147

Following are the major constraints considered in order to derive the trajectory parameters

- Attitude control is decoupled from guidance (to maintain pitch angle 50°) throughout phase
- Altimeters are operated and inputs are taken into navigation filter
- Phase duration is chosen sufficiently high compare to navigation filter convergence time (~2.5 s)
- Thrust level is chosen such that next phase has natural response under nominal condition

Fine Braking Phase

Major objective of fine braking phase is to bring velocity near zero to ensure hovering condition for next phase. Since fine braking phase is operated in close loop guidance, sensor aiding is not planned for inertial navigation system. Table 5 shows the detailed fine braking phase parameters

Table 5 Fine Braking Phase Parameters

Phase Parameters	
Duration, s	175 s ± 6 s
Guidance	3rd Order Polynomial Close Loop Guidance, Thrust regulation

Terminal constraint	Pitch Angle		0 ⁰	
	Velocity (Vh & Vv)		0 m/s & 0 m/s	
	Position (Dr & Alt.)		0 km & 800/1300 [m]	
	Acceleration		2g	
Initial Condition	Pitch Angle		50 ⁰	
	Velocity (Vh & Vv)		336 m/s & -58 m/s	
	Position (Dr & Alt.)		28.52 km & 6.88 km	
Navigation	Inertial Navigation (LIRAP Propagated) & Filter in background & updates from filter			
Initial Dispersion FB [#]	Radial, m	75*, 847**	Radial, m/s	1.711
	Along, m	590.927	Along, m/s	0.587
	Across, m	399.502	Across, m/s	1.147
FB End Dispersion [#]	Radial, m	252*,1113**	Radial, m/s	0.2*,2.8**
	Along, m	637.264	Along, m/s	0.3*,1**
	Across, m	540.271	Across, m/s	0.12*,1.45**
Initial Guidance Execution Error	Radial, [m]	847, 75	Radial,	0.8
	Along, [m]	80	Along,	0.5
	Across, [m]	40	Across,	0.4
Terminal Guidance Execution Error	Radial, [m]	4	Radial,	0.3
	Along, [m]	3	Along,	0.3
	Across, [m]	3	Across,	0.3

Following are the major constraints considered in order to derive the trajectory parameters

- Fine braking phase guidance should be able to correct for navigation error updated in attitude hold phase
- All four engines need to be operated, requirement of engine switch during fine braking phase needs to be avoided. 2g terminal acceleration is targeted to ensure engines are well within operating region and thrust continuity is ensured at next phase when engines are switched off
- In case there is no aiding inertial systems, altitude at the end of fine braking phase needs to be re-planned to a higher altitude in order to avoid pre-mature touch down
- In case state vectors are beyond guidance convergence limits, a new and novel real time retargeting algorithm is in place to adjust the landing site ensuring safe landing and soft landing
- If state vectors are much beyond the limits then emergency landing is planned where salvage guidance is invoked

Altitude targeted at the end of fine braking phase is based on additive error from various error sources

- Additive inertial navigation error for 175 [s] of pure propagation
- Initial aiding error at start of fine braking phase
- Terrain variation at landing site

Table 6 shows the error considered for designing the targeted altitude.

Table 6 Navigation error budgeting for various condition

Error Source	Value	Remark
Initial Navigation Error	75 [m]	In case altimeter aiding is successful
	850 [m]	In case of no altimeter aiding
Navigation Propagation Error	250 [m]	Based on Additive extreme simulation
Landing site Terrain Variation	50 [m]	Based on DEM statistics
Total Additive Error	375 [m]	Targeted altitude: 800 [m]
	1150 [m]	Targeted altitude: 1300 [m] (Contingency path)

Terminal Descent Phase

In terminal descent phase navigation frame of reference is switched from moon centre inertial to local reference frame. Terminal descent phase consist of multiple subphase i.e.

- Hovering #1 phase:** This phase is introduced to hover at higher altitude and initiate local navigation. During this phase, two engines are shutdown in order to avoid engine operating at lower thrust regime. This phase lasts based on time and timeline of activity done during this phase is shown in figure 4. At the end of phase, local navigation gets aided using altimeter and next phase is initiated. First 4 s is apportioned for engine transient settling followed by 8 s for local navigation filter settling.

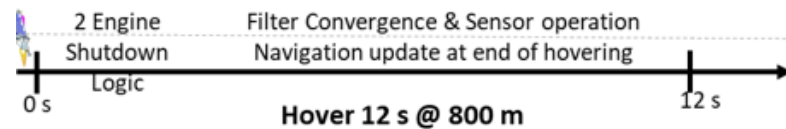


Figure 4 Timeline of operation during first hovering

- Vertical Descent #1 phase:** At the end of hovering #1, lander performs a descent to 150 m altitude. It is required to reach 150 [m] with a predefined accuracy for successful operation of LHDAC. Hence during this phase, navigation gets aided at every 10 s interval till 40 s before the elapse of the phase.
- Hovering #2 Phase:** This phase is meant for LHDAC camera. LHDAC camera gets operated after 6 s of hovering start. 6 s is allocated for attitude control settling to ensure stable platform for camera operation. Post LHDAC camera snap, lander waits for clue from hazard detection camera to initiate hazard avoidance maneuver. This phase continues till lander receives clue from LHDAC or timeout happens. Detailed timeline is shown in figure 5

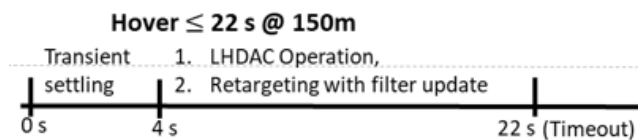


Figure 5 Timeline of operation during Hovering #2

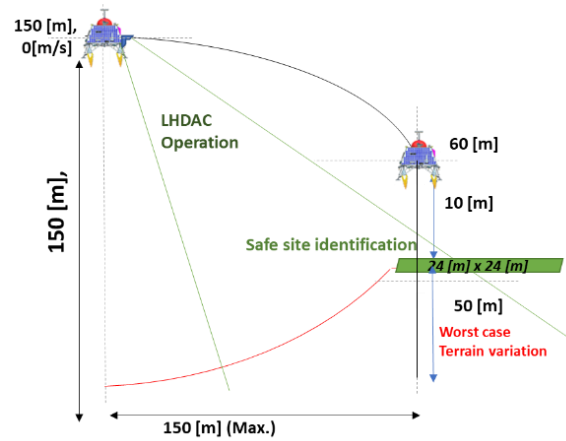


Figure 6 safe altitude targeting over identified landing site

- **Re-Targeting Phase:** Once Lander receives safe site co-ordinate from LHDAC, it does a parabolic retargeting to 60 m over safe site. During this phase, lander can cover up to 150 m downrange. This downrange travel can lead to terrain variation and can impact altimeter performance. Hence no aiding to navigation is planned during this phase. Maximum terrain variation for a horizontal travel of 150 m is 50 m. Hence 60 m above the safe site is chosen. Figure 6 shows pictorial representation of terrain variation
- **Vertical Descent #2 Phase:** Once Lander reaches over safe site, vertical descent to 10 m is planned. During this phase, navigation gets aided every 5 s interval till 20 s before the elapse of phase.
- **Constant Velocity Descent Phase:** Upon reaching to 10 m altitude over safe site, further no more navigation aiding is done. Lander velocity is controlled near to -1 [m/s] and final descent is performed until touchdown is detected through touchdown sensor or a timeout logic. In flight the touchdown through the touchdown sensor was confirmed.

Section IV Chandrayaan-3 Robustness and Multi-Mission Path

In Chandrayaan-2 mission, powered descent trajectory was designed with 20 [N] margin with respect to thrust dispersion. The design was fuel optimal with high sensitivity for inflight perturbation. During Chandrayaan-2 flight large dispersion in terms of thrust and state vector was observed during attitude hold phase. This lead the initial conditions of fine braking phase guidance to be way off than expected dispersion. Fine braking target milestone was constrained to a pin point landing site. This led to fine braking guidance convergence issues in flight.

To overcome such scenarios, multi mission path was introduced in Chandrayaan-3. Also, robustness in trajectory design was introduced to cater to large thrust dispersion in propulsion system. Following were the major modifications done in trajectory design and powered descent strategy to enhance robustness and fault tolerant to multiple failures

- Baseline trajectory was design with assumption of 40 [N] thrust shortfall in propulsion system which added robustness in terms of propulsion dispersion. Even though available thrust from throttleable engines are up to 790 [N], under nominal condition thrust demand from NGC will never exceed 750 [N].

- In fine braking phase, multi mission paths are introduced to cater for scenarios where system states are much beyond designed limits. Figure 7 shows the multi mission path introduced in fine braking phase to enhance robustness and fault tolerance to multiple failure scenario
 - At the end of attitude hold phase, a check is done to see if navigation is aided with altimeters. In case of no altimeter aiding (from both KaRa & LASA), targeted altitude at the end of fine braking phase is changed to 1300 [m] against 800 [m] nominal.
 - Guidance convergence check is done at the end of fine braking phase. Function is evaluated based on stored co-efficient onboard to see if states are within guidance limits (green region). In case states are beyond limit (as shown in orange region), target landing site is shifted appropriately to ensure states are within convergence limits (green region)
 - In case states are much beyond the limits, emergency landing is invoked. In emergency landing situation, only altitude and velocities are controlled while horizontal range is unconstrained to wider the range of convergence

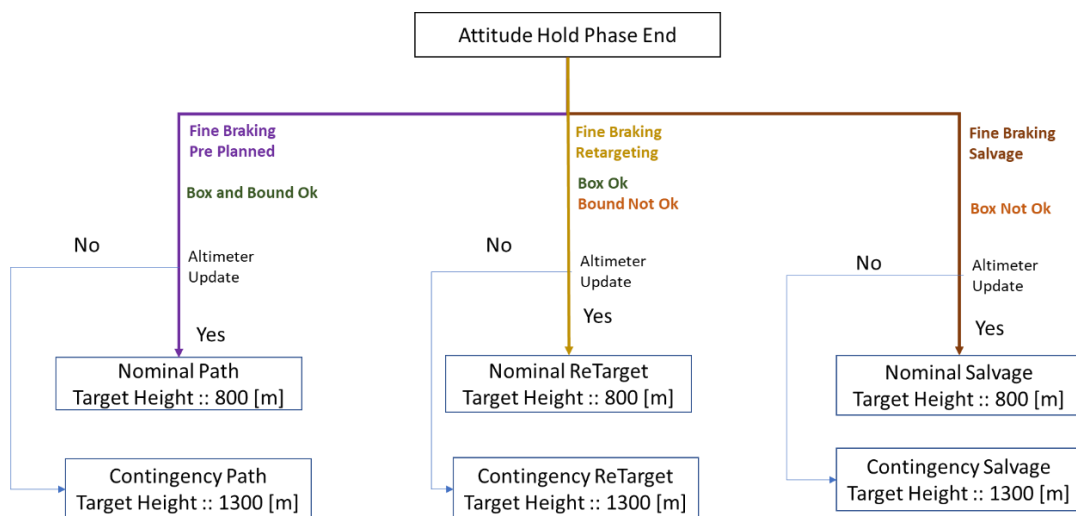
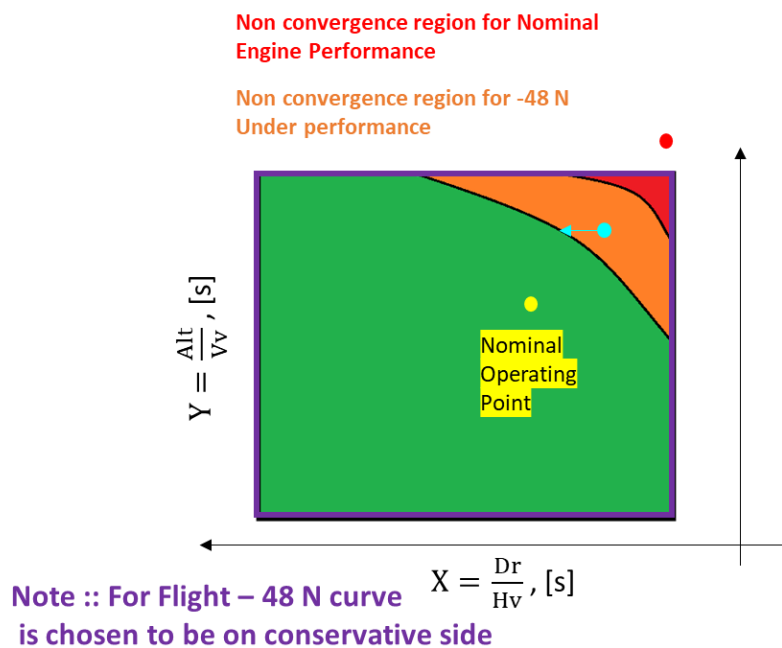


Figure 6 Multi mission path decision tree at attitude hold phase end

Section V Chandrayaan-3 Trajectory Simulation with inflight performance

Chandrayaan-3 made successful landing in the evening hours [18:04 IST] of August 23rd 2023. There was height update at the end of Attitude Hold phase leading to Fine braking with target of 800m as per nominal mission path. Hazard avoidance camera indicated that the safe site was directly below.

Nominal lander parameters and initial condition for powered descent start are shown in Table 7. Figure 7 and Figure 8 shows the altitude variation during flight powered descent phase when compared against nominal and expected upper and lower bound. Figure 9 and figure 10 shows vertical velocity during flight powered descent phase with upper and lower bound. Figure 11 and figure 12 shows thrust demanded by NGC during flight powered descent phase with upper and lower bound. Figure 13 and Figure 14 shows pitch angle variation during flight powered descent phase with upper and lower bound. Figure 15 shows Monte Carlo simulation statistics on landing error ellipse and observed landing error.

Table 7 Nominal simulation parameter

	Value
Lander Mass at Powered Descent Start	1729 [kg]
Altitude	30 [km]
Downrange	745 [km]
Horizontal Velocity	1688 [m/s]
Vertical Velocity	0.0 [m/s]
Thrust	4 x 800 [N]
ISP	312 [s]
Rough Braking Target	
Altitude	7.5 [km]
Downrange	31 [km]
Vertical Velocity	-61 [m/s]
Horizontal Velocity	360 [m/s]
Pitch Angle	50 ⁰
Terminal Thrust	4 x 740 [N]
Fine Braking Target	
Altitude	800 [m]
Downrange	0.0 [km]
Vertical Velocity	0.0 [m/s]
Horizontal Velocity	0.0 [m/s]
Pitch Angle	0 ⁰
Terminal Thrust	2g
Terminal Descent Phase	
Altitude	800 [m], 150 [m], 60 [m], 10 [m]
Vertical Velocity	0.0 [m/s]
Horizontal Velocity	0.0 [m/s]
Terminal Thrust	G
Pitch Angle	0 ⁰

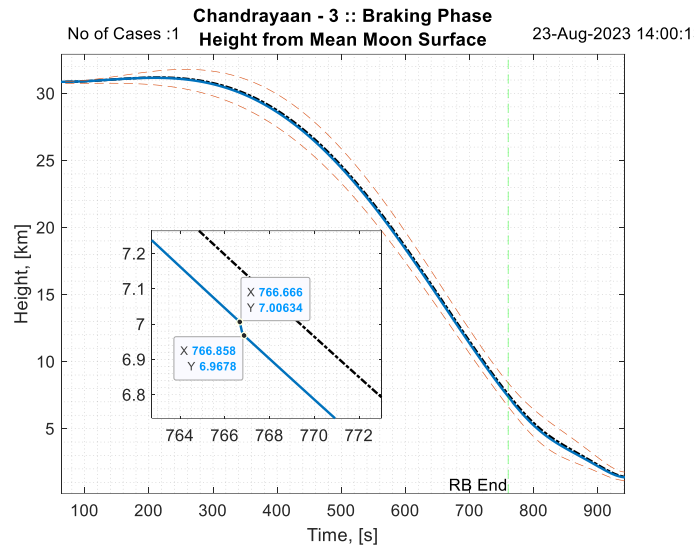


Figure 7 Altitude plot during braking phase

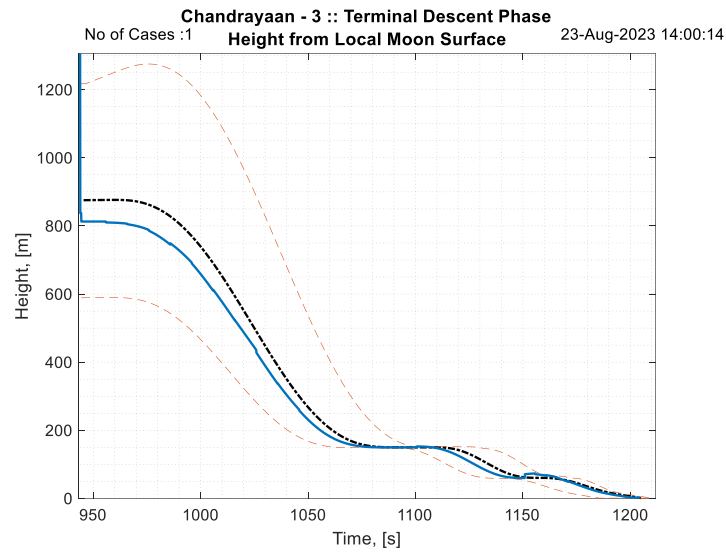


Figure 8 Altitude plot during terminal descent phase

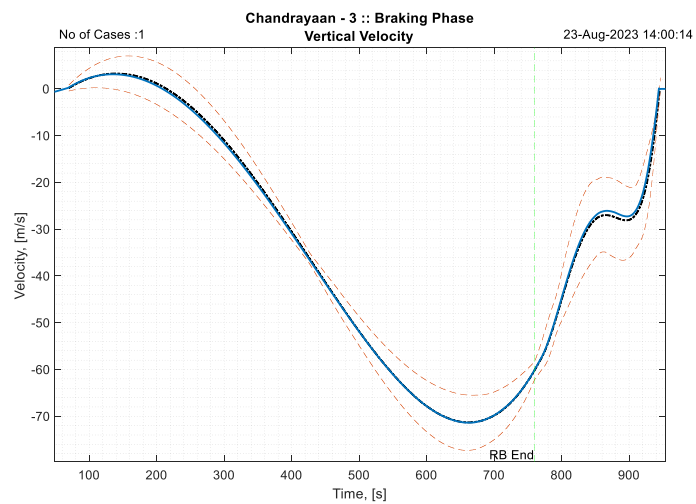


Figure 9 Vertical velocity during braking phase

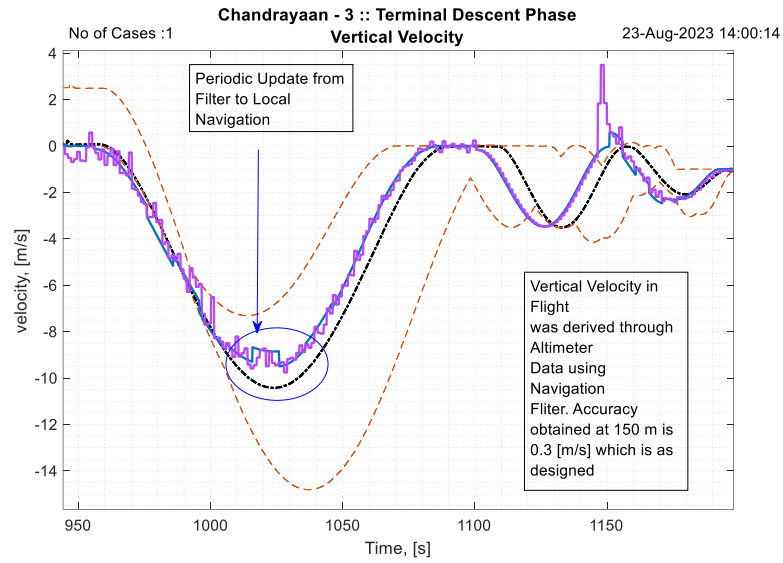


Figure 10 Vertical velocity during terminal descent phase

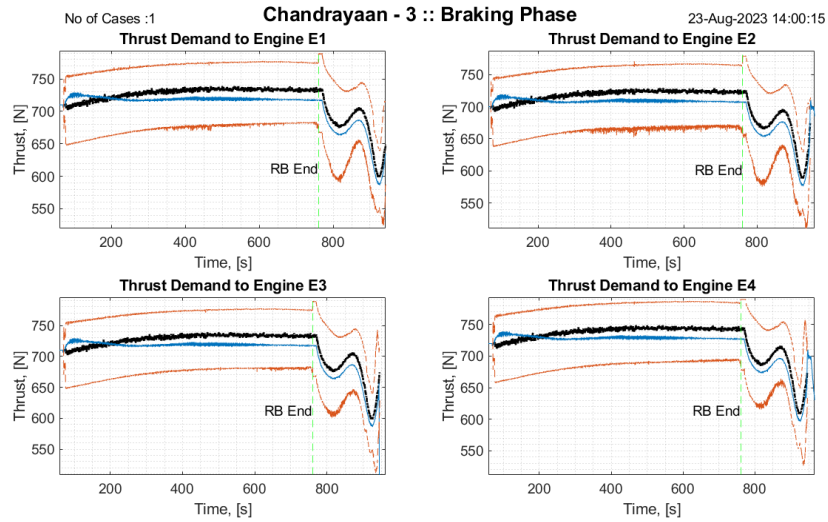


Figure 11 Thrust demand during braking phase

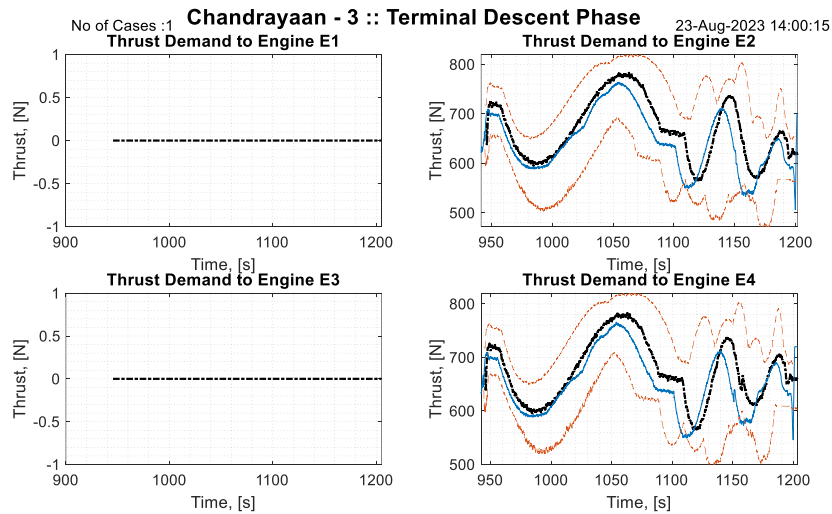


Figure 12 Thrust demand during terminal descent phase

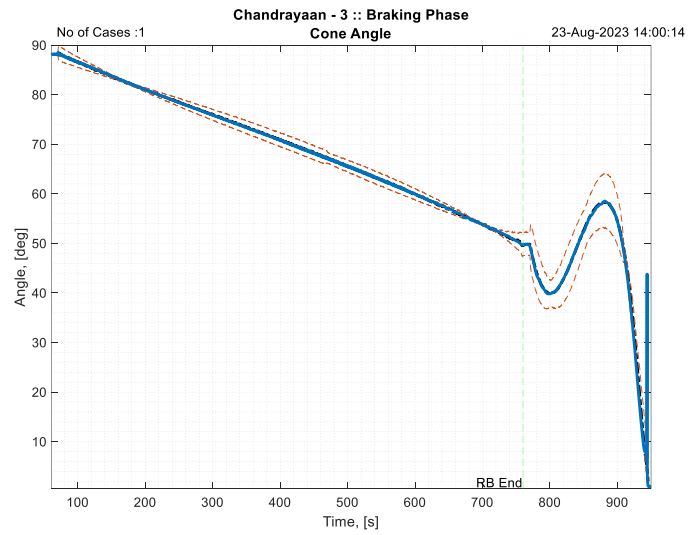


Figure 13 Pitch angle during terminal descent phase

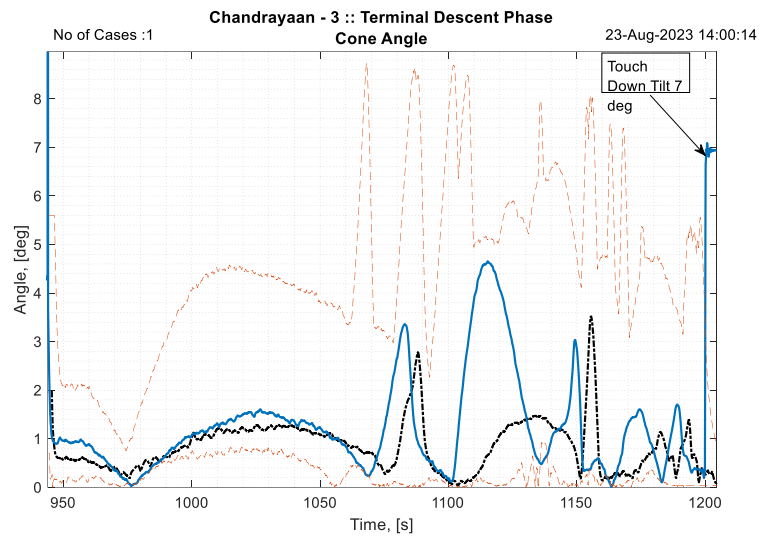


Figure 14 Pitch angle during terminal descent phase

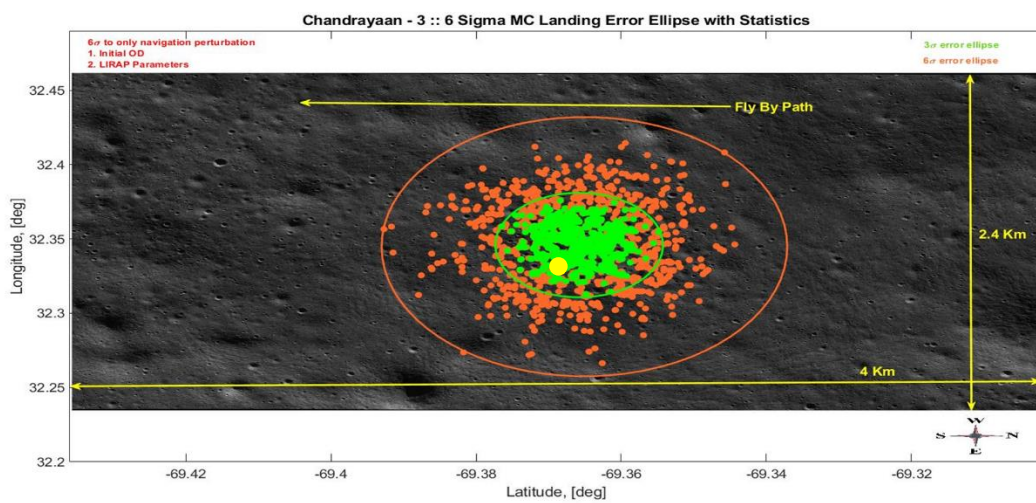


Figure 15 Chandrayaan-3 landing statistics for Monte Carlo simulation

Section VI Conclusion

Chandrayaan-3 trajectory was designed to ensure sufficient robustness with respect to propulsion system variation and state vector dispersion with continuity in the thrust profile commanded to engines. Multi-mission paths, mission critical decision logics and mitigation strategies were worked out to ensure robustness against various failure conditions like, no altimeter update in attitude hold phase, real time retargeting in fine braking phase and salvage situation under large propulsion system dispersion. In flight NGC system performance was well within $3\text{-}\sigma$ and nominal path got executed as a part of landing sequencer during powered descent. All system parameters observed in flight were nominal and well within design limits

Section VII Acknowledgement

Authors would like to express sincere gratitude to Chairman ISRO for his continuous support. Authors would also like to thank Mr. Sankaran M (Director URSC), Mr. Sudhakar S (Deputy Director, Controls and Digital Area) and Dr. Veeramuthuvel (Project Director, Chandrayaan - 3) for their guidance and suggestions during Chandrayaan-3 key reviews. Authors would also like to thank Dr. U P Rajeev (Group Director MSSG, VSSC), Mr. Jagannath Das (Group Director OSDG, Onboard Software Development Group) Dr. M S Siva (Group Director CDDG, URSC) and Dr. Ravi Kumar L (Group Director MSG, URSC) for their endless support. Authors would also like to thank the entire NGC team including onboard software, OILS, SILS, Hardware Simulator, HILS & Quality assurance team for their endless support during all major reviews.